

and, in some clusters, with sex and the presence of ornaments. Jomon subsistence, then, was not only broad-based but socially differentiated well before rice ever arrived (Kiriya and Kusaka 2017). Together, the zooarchaeological and isotopic evidence describes an economy of real flexibility: one that drew on whichever combination of shellfish, fish, terrestrial game, and wild plants was locally most reliable, among people who already ate somewhat different diets from one another.

Pinning down exactly when and how rice cultivation entered this system takes different methods, since carbonized grains and other plant remains survive far more poorly than shell or bone. Paleoethnobotanists lean heavily on impressions left by rice husks and grains in unfired clay, which are preserved permanently once the pottery is fired and can reveal a crop's presence even where the grains themselves have long since rotted away. Endo and Leipe's survey of seed impressions in Jomon and Yayoi pottery from across the archipelago found that rice impressions appear first in southwestern Japan and spread northeastward only gradually over the following centuries, with foxtail millet (*Setaria italica*) and broomcorn millet (*Panicum miliaceum*) serving as earlier, complementary crops in some regions before rice cultivation took over (Endo and Leipe 2022). The method has a real limit, one the authors acknowledge themselves: an impression records only that a plant was present at a site, not how much of a community's diet it supplied, which makes it better for tracing geography and timing than for measuring dietary importance. Chronological precision comes instead from radiocarbon dating performed directly on charred rice grains rather than on associated charcoal, avoiding the risk of dating fuel wood or other material of a different age than the crop itself. Leipe and colleagues' Bayesian modeling of these direct dates found that rice could have appeared as early as the eleventh century BCE in the Central Highlands of Honshu, likely used mainly for ritual purposes among Final Jomon